Chapter 3

Related Work

This chapter reviews the research that this thesis builds on and distinguishes it from existing approaches. The thesis lies at the intersection of three research areas (Figure 3.1): (i) large language models for medicine and patient education, (ii) knowledge graphs and neuro-symbolic reasoning for LLMs, and (iii) automatic prompt optimisation and the automatic generation of educational videos. Section 3.1 discusses medical LLMs and their limitations for patient education. Section 3.2 reviews approaches that ground medical LLMs in knowledge graphs, and Section 3.3 reviews methods that combine symbolic rules with LLM reasoning. Section 3.4 covers automatic prompt optimisation, and Section 3.5 covers the generation of educational videos. Section 3.6 compares all approaches and summarises the research gap addressed by this thesis.

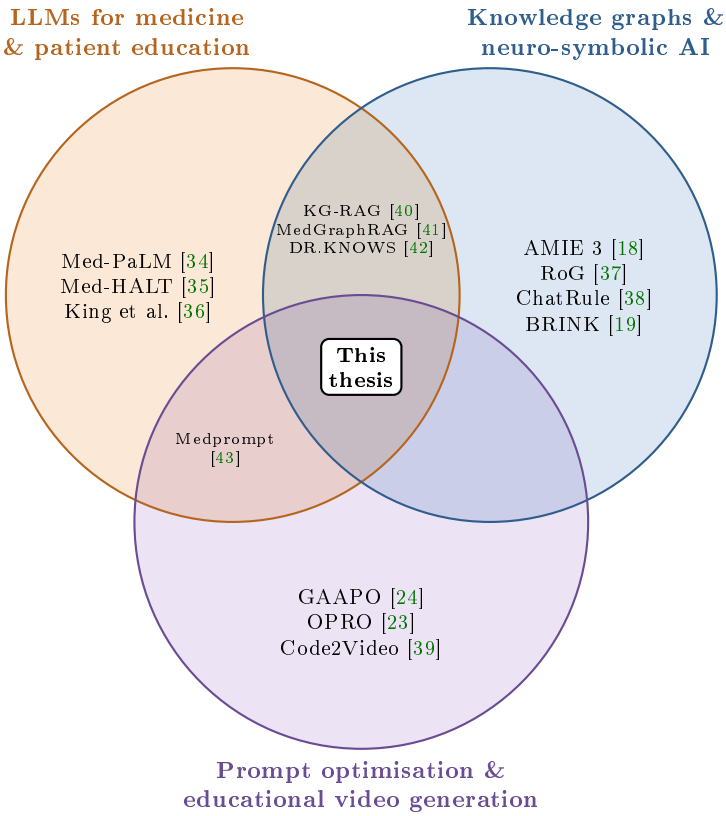


Figure 3.1: Positioning of this thesis. Existing work covers one or two of the three research areas; this thesis combines guideline-based knowledge graphs, rule-guided reasoning, prompt optimisation and patient-oriented storyboards for educational videos.

3.1 Large Language Models in Medicine and Patient Education

Medical question answering. Singhal et al. [34] introduced MultiMedQA, a benchmark that combines professional medical exams, research questions and consumer health questions, and adapted the general-purpose model Flan-PaLM to the medical domain through instruction prompt tuning. In a human evaluation, 92.6 % of the long-form answers of the resulting model, Med-PaLM, were judged to be in line with scientific consensus, compared with 92.9 % of clinician answers; nevertheless, the model remained inferior to clinicians on several other evaluation axes. Its successor Med-PaLM 2 [44] further narrowed this gap. Nori et al. [43] showed that a general model (GPT-4) can reach similar results without any medical fine-tuning when it is combined with a carefully designed prompting strategy (Section 3.4). Surveys of the field [13] conclude that LLMs are promising for clinical documentation, education and patient communication, but that their reliability, transparency and accountability are not yet sufficient for unsupervised use.

Patient communication and patient education. LLMs have also been evaluated directly on patient-facing tasks. Ayers et al. [45] compared answers by physicians and by ChatGPT to 195 patient questions from a public online forum; in 78.6 % of 585 evaluations, healthcare professionals preferred the chatbot answer, which was rated higher for both quality and empathy. In cardiology, King et al. [36] used GPT-4 to simplify 143 heart-failure patient education materials from leading US cardiology institutions. The median reading level decreased from grade 10.3 to grade 7.3, and a cardiologist rated all simplified texts as equally accurate as the originals. These studies show that LLMs can make medical information more accessible. However, they rely on general-purpose models without explicit medical knowledge, they evaluate plain text only, and correctness is checked manually after generation.

Hallucinations in medical LLMs. The main obstacle to the safe use of LLMs for patients is hallucination (Section 2.2.3). Pal et al. [35] proposed Med-HALT, a benchmark that tests both reasoning hallucinations (e.g. choosing an answer to a question with a false premise) and memory-based hallucinations (e.g. inventing references), and found that all evaluated LLMs were affected. Kim et al. [46] analysed eleven foundation models on seven medical tasks and, together with a survey of clinicians, concluded that medical hallucinations are frequent in practice. Interestingly, they also observed that chain-of-thought prompting reduced hallucinations in most comparisons, which supports the idea of making the reasoning of a model explicit.

> **Research Gap**
>
> Medical LLMs can produce fluent and empathetic answers, but they are not explicitly grounded in clinical guidelines, their reasoning is not verifiable, and existing patient-education studies focus on text. Approaches are needed that connect patient-oriented generation with explicit, guideline-based medical knowledge.

3.2 Knowledge-Graph-Enhanced LLMs in the Biomedical Domain

A common way to reduce hallucinations is to supply LLMs with external knowledge at inference time (Section 2.4.1). Xiong et al. [47] systematically benchmarked retrieval-augmented generation for medicine on MIRAGE, a collection of 7,663 questions from five medical QA

datasets. Their MedRAG toolkit improved the accuracy of six LLMs by up to 18 % compared with chain-of-thought prompting. These systems retrieve unstructured text passages, for example from PubMed or textbooks.

Several works retrieve structured knowledge from *knowledge graphs* instead. Soman et al. [40] proposed KG-RAG, which extracts disease-specific context from SPOKE, a large biomedical knowledge graph, and adds it to the prompt. By pruning the retrieved context with embeddings, they reduced the number of prompt tokens by more than 50 % while improving the answers of models such as Llama-2 and GPT-4. Gao et al. [42] presented DR.KNOWS, which selects paths from a UMLS-based knowledge graph that are relevant to a patient's electronic health record and uses them to support diagnosis prediction; clinicians judged the resulting diagnostic rationales to be clinically sound. Wu et al. [41] proposed MedGraphRAG, which links private medical documents to trusted medical sources and vocabularies in a three-level graph, and retrieves from this graph so that answers are supported by evidence. Outside medicine, Think-on-Graph [48] lets an LLM explore a knowledge graph step by step with beam search, and Microsoft's GraphRAG [49] builds a graph of entities and communities from text to answer global questions about a document collection.

All of these approaches use the knowledge graph only at *inference time* and assume that the facts needed to answer a question are present in the graph. Zhou et al. [19] questioned this assumption with the BRINK benchmark (Section 2.4.1): when the triple that directly supports an answer is removed, current KG-RAG methods show limited reasoning ability, often fall back on knowledge memorised during pre-training, and generalise differently depending on their design. This finding is highly relevant for medical knowledge graphs, which are incomplete by nature (Section 2.1.2).

Example 15: Retrieval versus learnt reasoning under missing knowledge

Assume the guideline knowledge graph contains the facts *(Drug X, belongsToClass, MRA)* and *(MRA, recommendedFor, HFrEF)*, but *not* the direct fact *(Drug X, recommendedFor, HFrEF)*. A patient asks: *"Is Drug X recommended for my type of heart failure?"*

- A **retrieval-only** system searches for the direct triple, does not find it, and either answers "no information" or falls back on what the LLM memorised during pre-training.
- A model that was **fine-tuned on rule-based reasoning chains**, e.g. derived from the rule $belongsToClass(x, c) \wedge recommendedFor(c, d) \Rightarrow recommendedFor(x, d)$, has learnt to combine the two remaining facts and can answer: *"Drug X is an MRA, and MRAs are recommended for HFrEF; therefore ..."*, while showing the facts it used.

This illustrates why this thesis combines knowledge-graph context at inference time with reasoning patterns learnt from symbolic rules.

Research Gap

Biomedical KG-enhanced LLMs ground answers in retrieved facts but do not teach the model how to *reason* over the graph when facts are missing. Moreover, they target clinicians or benchmark questions, not patient education based on clinical practice guidelines and patient questionnaires.

3.3 Symbolic Rules and Neuro-Symbolic Reasoning with LLMs

Rule learning over knowledge graphs. Learning logical rules from knowledge graphs has a long tradition. The AMIE family [16, 17, 18] mines closed Horn rules under the Partial Completeness Assumption and is the rule miner used in this thesis (Section 2.3.1). AnyBURL [50] learns rules bottom-up by sampling paths from the graph and generalising them, which makes it efficient for knowledge-graph completion. Neural approaches such as Neural LP [51] learn rules in a differentiable way, trading some interpretability for end-to-end training. In all of these works, rules are used to *complete* the knowledge graph, not to train a language model.

LLMs and rules. More recent work connects rules with LLMs. ChatRule [38] uses an LLM to *generate* candidate rules from the semantic and structural information of a knowledge graph and ranks them with measures computed on the existing facts. Reasoning on Graphs (RoG) [37] fine-tunes an LLM to first generate relation paths that are grounded in the knowledge graph ("planning") and then to reason over the retrieved paths, producing faithful and interpretable answers. Closest to this thesis is Chain-of-Knowledge [52], which mines rules from knowledge graphs, turns them into a knowledge-reasoning dataset (KnowReason), and fine-tunes LLMs on it; to prevent the model from simply memorising the rules, the authors add a trial-and-error mechanism.

Reasoning chains as training data. Training models on explicit reasoning chains has been shown to be effective in general. Hsieh et al. [53] demonstrated that smaller models fine-tuned on rationales extracted from a large LLM can outperform much larger models while using less training data. Rule groundings from a knowledge graph provide such rationales without a teacher LLM, and their correctness can be traced back to explicit facts (Section 2.3.2).

> **Research Gap**
>
> Neuro-symbolic approaches that combine rules and LLMs have so far been evaluated on open-domain benchmark knowledge graphs (e.g. Freebase-based question-answering datasets). It is unclear whether rule-guided chain-of-thought fine-tuning improves LLM performance in the medical domain, where knowledge comes from heterogeneous sources (guidelines, terminologies and patient questionnaires) and where rules must be filtered by confidence before they are used.

3.4 Automatic Prompt Optimisation

Since the quality of LLM outputs depends strongly on the prompt, several methods search for better prompts automatically. Automatic Prompt Engineer (APE) [54] lets an LLM propose candidate instructions from input–output examples and selects the best one by its score. ProTeGi (APO) [22] improves prompts with natural-language "gradients", and OPRO [23] uses an LLM as an optimiser that sees the history of previous prompts and their scores. Evolutionary approaches maintain a population of prompts: EvoPrompt [55] uses an LLM to perform mutation and crossover and improved over human-written prompts on 31 datasets, and Promptbreeder [56] additionally evolves the mutation prompts themselves. GAAPO [24] combines mutation, crossover, few-shot augmentation, APO and OPRO in a single genetic algorithm and records which generator produced each improvement (Section 2.5).

In medicine, Medprompt [43] combined dynamic few-shot selection, self-generated chain-of-thought explanations and choice-shuffle ensembling, which allowed GPT-4 to reach 90.2 % accuracy on the MedQA benchmark without any domain-specific fine-tuning. Medprompt

is a fixed composition of prompting techniques, not a search over prompts, and like most prompt-optimisation methods it was evaluated on multiple-choice questions.

> **Research Gap**
>
> Prompt-optimisation methods are mostly evaluated on classification and multiple-choice tasks with very large models. They have not been studied for generating *structured* patient-education content, such as video storyboards, with a small, domain-fine-tuned model whose medical content must not be changed by the optimised prompt.

3.5 Automatic Generation of Educational Videos

Video is an effective format for patient education: a systematic review found strong evidence that video-based interventions improve the knowledge of patients with chronic illnesses [32]. Producing such videos, however, requires medical experts, script writers and designers. Recent work therefore investigates how LLMs can automate parts of this process. TheoremExplainAgent [57] uses LLM agents to plan and write Manim animation code for explanation videos of mathematical and scientific theorems; the best configuration produced videos for 93.8 % of 240 theorems, although most videos still contained minor visual layout problems. Code2Video [39] also generates educational videos as executable code, using a planner, a coder and a critic agent, and reports a 40 % improvement over direct code generation. In healthcare, Haider et al. [58] compared AI-generated avatar videos with text handouts for postoperative education in a pilot randomised trial. Knowledge scores did not differ significantly, but participants rated the videos as clearer and more engaging.

These systems focus on the *production* of a video from a given topic and rely on the general knowledge of the LLM for its content. They do not verify the medical statements against a trusted knowledge source, and they do not adapt the content to the needs of a specific patient population.

> **Research Gap**
>
> LLM-based video generation has been studied mainly for STEM topics, while medical studies use generic AI tools without knowledge grounding. A method is missing that produces patient-oriented video storyboards whose medical content is derived from guideline-based knowledge and whose structure can be validated before production.

3.6 Summary and Research Gap

Table 3.1 compares representative approaches along the dimensions that are relevant for this thesis. Each line of research covers only part of the problem: medical LLMs and patient-education studies provide patient-oriented language but no explicit knowledge; biomedical KG-RAG systems ground answers in knowledge graphs but only at inference time; neuro-symbolic methods learn from rules but have not been applied to medicine; prompt optimisation and video generation have not been combined with knowledge-grounded medical content.

Based on this analysis, the research gap addressed in this thesis can be summarised as follows.

> **Research Gap of this Thesis**
>
> There is no approach that (i) represents clinical guidelines and patient questionnaires about cardiovascular disease as knowledge graphs, (ii) turns symbolic rules mined from

Approach	Med.	KG	Rules	CoT-FT	P-Opt	Patient	Video
Med-PaLM [34]	✓	–	–	–	–	(✓)	–
King et al. [36]	✓	–	–	–	–	✓	–
MedRAG [47]	✓	–	–	–	–	–	–
KG-RAG [40]	✓	✓	–	–	–	–	–
DR.KNOWS [42]	✓	✓	–	–	–	–	–
MedGraphRAG [41]	✓	✓	–	–	–	–	–
RoG [37]	–	✓	(✓)	(✓)	–	–	–
ChatRule [38]	–	✓	✓	–	–	–	–
Chain-of-Knowledge [52]	–	✓	✓	✓	–	–	–
Medprompt [43]	✓	–	–	–	(✓)	–	–
GAAPO [24]	–	–	–	–	✓	–	–
TheoremExplainAgent [57]	–	–	–	–	–	–	✓
Code2Video [39]	–	–	–	–	–	–	✓
This thesis	✓	✓	✓	✓	✓	✓	✓

Table 3.1: Comparison of related approaches. **Med.**: medical domain; **KG**: uses a knowledge graph; **Rules**: uses symbolic rules; **CoT-FT**: fine-tunes the LLM on (rule-based) reasoning chains; **P-Opt**: automatic prompt optimisation; **Patient**: patient-oriented output; **Video**: educational video or storyboard generation. ✓ = yes, (✓) = partially, – = no. For this thesis, **Video** refers to storyboard scripts as input for video production.

these graphs into chain-of-thought training data for a medical LLM, (iii) evaluates whether this improves reasoning when knowledge is incomplete, and (iv) uses the resulting model, together with an automatically optimised prompt, to generate patient-oriented storyboards for educational videos. This thesis addresses this gap with the semantic-driven approach presented in Chapter 4.

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